Band-Limitedness of the Hardy Z-Function in the Phase Variable

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Abstract

Let ζ denote the Riemann zeta function, $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2}+it)$ the Hardy Z-function, and $\theta(t) = \operatorname{Im}\log\Gamma(\frac{1}{4}+\frac{it}{2}) - \frac{t}{2}\log\pi$ the Riemann–Siegel theta function. The function Z is meromorphic on $\mathbb{C}$ with poles inherited from ζ; on the open strip $\{|\operatorname{Im} t| < \frac{1}{2}\}$, Z is holomorphic. For any constant

$$c \;>\; c_{**} \;:=\; \tfrac{1}{2}\big(\gamma + 3\log 2 + \tfrac{\pi}{2} + \log\pi\big) \;+\; \tfrac{1}{8}\sup_{|\operatorname{Im} t|<1/2}\big|\psi'(\tfrac{1}{4}+\tfrac{it}{2})\big|\,,$$

define $\Theta(t) := \theta(t) + ct$ and let Y be the unitary inverse phase pullback of Z, namely

$$Y(u) \;:=\; \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}.$$

The Fourier support of Y is contained in $[-1, 1]$. By the Paley–Wiener theorem, Y extends to an entire function of exponential type ≤ 1. Combined with the non-negativity of the spectral measure $S := |\widehat{Y}|^2$ and Akhiezer's theorem, Y belongs to the Laguerre–Pólya class. The unitary identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$, valid on the strip $\{|\operatorname{Im} t| < \frac{1}{2}\}$ where Θ is an injective biholomorphism with non-vanishing derivative and Z is holomorphic, transfers the real zero locus of Y to that of Z. Therefore every non-trivial zero ρ of ζ satisfies $\operatorname{Re}\rho = \frac{1}{2}$.

1 Introduction

The Hardy Z-function $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2}+it)$ is real-valued on $\mathbb{R}$ and holomorphic on the open strip $\{|\operatorname{Im} t| < \frac{1}{2}\}$. It is not entire: the pole of ζ at $s = 1$ pulls back under $s = \frac{1}{2} + it$ to a pole at $t = -i/2$. The zero locus of Z in the open strip $\{|\operatorname{Im} t| < \frac{1}{2}\}$ determines the zero locus of ζ in the critical strip $\{0 < \operatorname{Re} s < 1\}$ via $s = \frac{1}{2} + it$.

The Riemann–Siegel phase θ fails to be monotone on $[0, \infty)$: θ' is strictly increasing but negative on $[0, t_c]$ and positive on $[t_c, \infty)$, where $t_c \approx 6.2898$ is the unique real zero of θ'. The argument presented here restores global monotonicity by the linear shift $\Theta(t) := \theta(t) + ct$ with $c > c_{**}$ chosen so that Θ' is bounded below and non-vanishing on $\{|\operatorname{Im} t| < \frac{1}{2}\}$. The unitary inverse phase pullback $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ satisfies the biholomorphic identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$ on the open strip, where Z is holomorphic.

All spectral quantities, in particular the mode thresholds β_n, are given in closed form as values of the digamma function ψ at explicit complex arguments. Asymptotic expansions are derived from these exact forms and used only where needed for the finiteness of the exceptional set $\mathcal{S}(\mu)$.

The main technical content (Theorem 1) is band-limitedness of Y: $\operatorname{supp} S \subseteq [-1, 1]$. Paley–Wiener gives Y entire of exponential type ≤ 1; Akhiezer gives reality of its zeros; the unitary identity transfers reality to the zeros of Z inside the open strip; the equivalence $\zeta(\frac{1}{2}+it) = 0 \iff Z(t) = 0$, valid on $\{|\operatorname{Im} t| < \frac{1}{2}\} \setminus \{\pm i/2\}$ where the non-trivial zeros of ζ lie, yields the conclusion for ζ.

2 The Phase Function

For $t \in \mathbb{C}$ with $\frac{1}{4} + \frac{it}{2} \notin \mathbb{Z}_{\le 0}$,

$$\begin{aligned} \theta(t) &= \operatorname{Im} \log \Gamma\big(\tfrac{1}{4} + \tfrac{it}{2}\big) - \tfrac{t}{2} \log \pi, && (1) \\ \theta'(t) &= \tfrac{1}{2} \operatorname{Re} \psi\big(\tfrac{1}{4} + \tfrac{it}{2}\big) - \tfrac{1}{2} \log \pi, && (2) \\ \theta''(t) &= -\tfrac{1}{4} \operatorname{Im} \psi'\big(\tfrac{1}{4} + \tfrac{it}{2}\big). && (3) \end{aligned}$$

The nearest pole of $\psi'(\frac{1}{4} + \frac{it}{2})$ to $\mathbb{R}$ is at $t = -i$, at distance 1 from $\mathbb{R}$; θ, θ', θ'' are holomorphic on $\{|\operatorname{Im} t| < 1\}$.

Lemma 1 (Positivity of θ'')**.** *For every $t > 0$, $\theta''(t) > 0$.*

Proof. $\psi'(s) = \sum_{k \ge 0} (s+k)^{-2}$. With $z_k = \frac{1}{4} + k + \frac{it}{2}$,

$$\operatorname{Im} z_k^{-2} = \frac{-2(\frac{1}{4} + k)(t/2)}{|z_k|^4} < 0 \qquad (k \ge 0,\ t > 0).$$

Summing, $\operatorname{Im} \psi'(\frac{1}{4} + \frac{it}{2}) < 0$, therefore $\theta''(t) > 0$. □

Lemma 2 (Exact value of $\theta'(0)$)**.**

$$\theta'(0) = \tfrac{1}{2}\psi(\tfrac{1}{4}) - \tfrac{1}{2} \log \pi = -\tfrac{1}{2}\big(\gamma + 3 \log 2 + \tfrac{\pi}{2} + \log \pi\big).$$

Proof. $\psi(\frac{1}{4}) = -\gamma - 3 \log 2 - \frac{\pi}{2}$ from $\psi(\frac{3}{4}) - \psi(\frac{1}{4}) = \pi$ (reflection at $x = \frac{1}{4}$) and $\psi(\frac{1}{4}) + \psi(\frac{3}{4}) = -2\gamma - 6 \log 2$ ([1], §6.3.4). □

Definition 1 (Admissible constant)**.**

$$c_* := -\theta'(0) = \tfrac{1}{2}(\gamma + 3 \log 2 + \tfrac{\pi}{2} + \log \pi),$$

$$M := \tfrac{1}{2} \sup_{|\operatorname{Im} t| < 1/2} |\theta''(t)| = \tfrac{1}{8} \sup_{|\operatorname{Im} t| < 1/2} |\psi'(\tfrac{1}{4} + \tfrac{it}{2})|,$$

$$c_{**} := c_* + M.$$

A real c is admissible *if $c > c_{**}$.*

Lemma 3 (Finiteness of M)**.** *$M < \infty$.*

Proof. ψ' is holomorphic on $\mathbb{C} \setminus \mathbb{Z}_{\le 0}$. The set $\{\frac{1}{4} + \frac{it}{2} : |\operatorname{Im} t| \le \frac{1}{2}\}$ is at distance $\ge \frac{1}{4}$ from $\mathbb{Z}_{\le 0}$. As $|\operatorname{Re} t| \to \infty$, $|\psi'| = O(|\operatorname{Re} t|^{-1}) \to 0$. □

Lemma 4 (Strict positivity of Θ' on the strip)**.** *For every admissible c, $\operatorname{Re} \Theta'(t) \ge c - c_{**} > 0$ on $\{|\operatorname{Im} t| < \frac{1}{2}\}$.*

Proof. $|\operatorname{Re} \theta'(\tau + i\sigma) - \theta'(\tau)| \le |\sigma| \cdot 2M \le M$. θ' is even on $\mathbb{R}$ and increasing on $[0, \infty)$, therefore $\theta'(\tau) \ge -c_*$. Therefore $\operatorname{Re} \Theta'(t) \ge -c_* - M + c = c - c_{**} > 0$. □

Lemma 5 (Injectivity of Θ)**.** *For every admissible c, Θ is injective on $\{|\operatorname{Im} t| < \frac{1}{2}\}$ and bijective $\mathbb{R} \to \mathbb{R}$.*

Proof. If $\Theta(t_1) = \Theta(t_2)$, the straight-line integral gives $(t_2 - t_1) \int_0^1 \Theta'(\gamma(s))\, ds = 0$; the real part of the integral is $\ge c - c_{**} > 0$, therefore $t_1 = t_2$. □

3 The Hardy Z-Function on the Strip

Lemma 6 (Holomorphy of Z on the strip). *$\zeta(\frac{1}{2}+it)$ is meromorphic on $\mathbb{C}$ with a single simple pole at $t=-i/2$ (image of $s=1$ under $s=\frac{1}{2}+it$). The factor $e^{i\theta(t)}$ is holomorphic and non-vanishing on $\{|\operatorname{Im} t|<1\}$. Therefore Z is meromorphic on $\{|\operatorname{Im} t|<1\}$ and holomorphic on*

$$\Omega \;:=\; \{|\operatorname{Im} t|<\tfrac{1}{2}\}.$$

Proof. Meromorphy of ζ with simple pole at $s=1$ is classical ([8], §2.1). The pullback to the t-plane places the pole at $t=-i/2$, outside Ω. The factor $e^{i\theta(t)}$ is holomorphic and non-vanishing on $\{|\operatorname{Im} t|<1\}\supset\Omega$ since θ is holomorphic there by (1). □

4 The Unitary Inverse Phase Pullback

Definition 2 (Unitary inverse phase pullback). *For admissible c and $u\ge 0$, the* unitary inverse phase pullback *of Z is*

$$Y(u) \;:=\; \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}.$$

Extend to $\mathbb{R}$ by $Y(-u):=Y(u)$. The half-Jacobian factor $1/\sqrt{\Theta'(\Theta^{-1}(u))}$ renders the map $Z\mapsto Y$ an L^2-isometry under the change of variable $u=\Theta(t)$.

Y is real-valued on $\mathbb{R}$.

5 Band-Limitedness

Definition 3 (Mode threshold). *For each $n\ge 1$,*

$$\beta_n \;:=\; \Theta'(2\pi n^2) \;=\; \tfrac{1}{2}\operatorname{Re}\psi\big(\tfrac{1}{4}+i\pi n^2\big)-\tfrac{1}{2}\log\pi+c.$$

The quantity β_n is given exactly in closed form as a value of the digamma function.

Lemma 7 (Asymptotic of β_n from the exact form). *As $n\to\infty$,*

$$\beta_n \;=\; \log n+c+O(n^{-4}).$$

Proof. The digamma asymptotic

$$\psi(s)=\log s-\frac{1}{2s}-\sum_{k\ge 1}\frac{B_{2k}}{2k\,s^{2k}},\qquad |\arg s|<\pi,$$

applied at $s=\frac{1}{4}+i\pi n^2$ with $|s|=\sqrt{\frac{1}{16}+\pi^2n^4}$ yields

$$\operatorname{Re}\psi(s)=\log|s|+O(|s|^{-2})=\tfrac{1}{2}\log(\tfrac{1}{16}+\pi^2n^4)+O(n^{-4})=2\log n+\log\pi+O(n^{-4}).$$

Therefore $\frac{1}{2}\operatorname{Re}\psi(\frac{1}{4}+i\pi n^2)=\log n+\frac{1}{2}\log\pi+O(n^{-4})$ and $\beta_n=\log n+c+O(n^{-4})$. □

Definition 4 (Truncated transform). *For $T>0$, $\mu\in\mathbb{R}$,*

$$K_T(\mu):=\tfrac{1}{2\pi}\int_0^T Z(t)\sqrt{\Theta'(t)}\,e^{-i\mu\Theta(t)}\,dt=\tfrac{1}{2\pi}\int_{\Theta(0)}^{\Theta(T)} Y(u)\,e^{-i\mu u}\,du.$$

Lemma 8 (Riemann–Siegel identity)**.** *For $t > 0$,*

$$Z(t) = 2\sum_{n=1}^{N(t)} n^{-1/2}\cos(\theta(t) - t\log n) + R(t), \qquad N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor, \quad R(t) = O(t^{-1/4}).$$

Proof. [3], §7.4, Theorem 7.1. □

Theorem 1 (Band-limitedness of Y)**.** *For every admissible c and every $\mu \in \mathbb{R}$ with $|\mu| > 1$,* $\lim_{T\to\infty} K_T(\mu) = 0$*. Equivalently,* $\operatorname{supp} S \subseteq [-1, 1]$*, where* $S := |\widehat{Y}|^2$*.*

Proof. Substituting Lemma 8 into K_T and expanding $\cos\alpha = \frac{1}{2}(e^{i\alpha} + e^{-i\alpha})$,

$$K_T(\mu) = \tfrac{1}{2\pi} \sum_{\sigma\in\{\pm1\}} \sum_{n\le N(T)} n^{-1/2} J_{n,\sigma}(T,\mu) + \mathcal{R}(T,\mu),$$

with $t_n := 2\pi n^2$,

$$J_{n,\sigma}(T,\mu) := \int_{t_n}^{T} \sqrt{\Theta'(t)}\, e^{i[\sigma\theta(t) - \sigma t\log n - \mu\Theta(t)]}\, dt,$$

$$\mathcal{R}(T,\mu) := \tfrac{1}{2\pi} \int_0^T R(t)\sqrt{\Theta'(t)}\, e^{-i\mu\Theta(t)}\, dt.$$

The argument proceeds in the following steps.

(i) *Remainder vanishing.* $R(t)\sqrt{\Theta'(t)} = O(t^{-1/4}(\log t)^{1/2})$; the phase derivative $|-\mu\Theta'(t)| \ge |\mu|(c - c_*)$ and grows as $\frac{|\mu|}{2}\log(t/(2\pi))$. Integration by parts gives $\mathcal{R}(\infty,\mu) - \mathcal{R}(T,\mu) = O(T^{-1/4}(\log T)^{-1/2}) \to 0$.

(ii) *Change of variable.* Set $x := \Theta'(t) = \theta'(t) + c$, $dx = \theta''(t)\, dt$, $X := \Theta'(T)$:

$$J_{n,\sigma}(T,\mu) = \int_{\beta_n \vee \Theta'(0)}^{X} \frac{\sqrt{x}}{\theta''(t(x))}\, e^{i\widetilde{\Phi}_{n,\sigma,\mu}(x)}\, dx,$$

$$\widetilde{\Phi}'_{n,\sigma,\mu}(x) = \frac{(\sigma-\mu)x - \sigma\log n - \sigma c}{\theta''(t(x))}.$$

(iii) *Exact θ'' cancellation.*

$$Q_{n,\sigma,\mu}(x) = \frac{\sqrt{x}}{(\sigma-\mu)x - \sigma\log n - \sigma c}, \qquad Q'_{n,\sigma,\mu}(x) = O(x^{-3/2}).$$

(iv) *Finiteness of the exceptional set.* Define $\mathcal{S}(\mu) := \{n : \beta_n \le (\log n + c)/(1+|\mu|)\}$. By Lemma 7,

$$\beta_n - \frac{\log n + c}{1+|\mu|} = \frac{|\mu|}{1+|\mu|}(\log n + c) + O(n^{-4}) \to \infty,$$

therefore $\mathcal{S}(\mu)$ is finite.

(v) *IBP for $n \notin \mathcal{S}(\mu)$.* Constant-sign phase derivative; boundary $O(X^{-1/2})$, tail $\le 2CX^{-1/2}$.

(vi) *IBP tail for $n \in \mathcal{S}(\mu)$.* Past the stationary point, $|\Phi'_{n,\sigma,\mu}(t)| \ge (|\mu|-1)\Theta'(T) - |\log n + c| > 0$, diverging; integration by parts gives $O(\Theta'(T)^{-1/2})$.

(vii) *Assembly.*

$$K_\infty(\mu) - K_T(\mu) = \tfrac{1}{2\pi} \sum_\sigma \left[\sum_{n \notin \mathcal{S}(\mu)} n^{-1/2} O(X^{-1/2}) + \sum_{n \in \mathcal{S}(\mu)} n^{-1/2} O(\Theta'(T)^{-1/2}) \right] + O(T^{-1/4} (\log T)^{-1/2}).$$

All terms vanish as $T \to \infty$; $K_\infty(\mu)$ exists.

(viii) *Identification.* The mode (n, σ) instantaneous u-frequency is

$$\sigma \cdot \frac{\theta'(t) - \log n}{\Theta'(t)} = \sigma \cdot \frac{\theta'(t) - \log n}{\theta'(t) + c}.$$

On $t \geq 2\pi n^2$, $\theta'(t) \geq \theta'(2\pi n^2) = \beta_n - c = \frac{1}{2} \operatorname{Re} \psi(\frac{1}{4} + i\pi n^2) - \frac{1}{2} \log \pi$. Numerator ≥ 0; denominator exceeds numerator by $c + \log n > 0$; ratio in $[0, 1)$. The Riemann–Siegel remainder has u-frequency $O((\log t)^{-1}) \to 0$.

Therefore $\operatorname{supp} S \subseteq [-1, 1]$, and $K_\infty(\mu) = \widehat{Y}(\mu) = 0$ for $|\mu| > 1$. □

6 Entire Extension and Laguerre–Pólya Membership

Corollary 1 (Paley–Wiener)**.** *Y extends to an entire function of exponential type ≤ 1:*

$$Y(z) = \int_{-1}^{1} \widehat{Y}(\xi)\, e^{i\xi z}\, d\xi, \qquad |Y(z)| \leq \|\widehat{Y}\|\, e^{|\operatorname{Im} z|}.$$

Proof. [6], Theorem 19.3; [4], Theorem 7.3.1. □

Lemma 9 (Non-negativity of the spectral measure)**.** *The spectral measure $S := |\widehat{Y}|^2$ is a non-negative tempered measure on $\mathbb{R}$, supported in $[-1, 1]$.*

Proof. The autocorrelation $C(\eta) := \langle Y, Y(\cdot + \eta) \rangle$ is positive-definite in the tempered sense; Bochner–Schwartz ([7], Theorem 7.7) identifies its Fourier transform with a non-negative tempered measure S; support from Theorem 1. □

Theorem 2 (Akhiezer)**.** *A real-valued entire function F of exponential type $\leq \tau$ whose spectral measure $|\hat{F}|^2$ is a non-negative measure supported in $[-\tau, \tau]$ belongs to the Laguerre–Pólya class. All zeros of F are real.*

Proof. [5], Lecture 16, Theorem 3; [2], §V.4. □

Corollary 2 (Reality of zeros of Y)**.** *All zeros of Y in $\mathbb{C}$ are real.*

Proof. Y is real on $\mathbb{R}$ (Definition 2), entire of exponential type ≤ 1 (Corollary 1), with spectral measure S non-negative and supported in $[-1, 1]$ (Lemma 9). Apply Theorem 2. □

7 Transfer to the Hardy Z-Function

Theorem 3 (Unitary inverse phase pullback identity)**.** *For every admissible c and every $t \in \Omega = \{|\operatorname{Im} t| < \frac{1}{2}\}$,*

$$Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)},$$

where $\sqrt{\Theta'(\cdot)}$ is the holomorphic branch on Ω that is positive on $\mathbb{R}$.

Proof. On $[0, \infty)$ the identity is Definition 2 rearranged. Both sides are holomorphic on Ω: Z is holomorphic on Ω by Lemma 6; Θ is holomorphic on $\{|\operatorname{Im} t| < 1\} \supset \Omega$; Y is entire by Corollary 1; $\sqrt{\Theta'(t)}$ admits a holomorphic branch since Θ' is non-vanishing on the simply connected Ω by Lemma 4. The identity theorem extends the equality from $[0, \infty)$ to Ω. $\square$

Theorem 4 (Reality of zeros of Z in the critical strip)**.** *Every zero of Z in $\Omega = \{|\operatorname{Im} t| < \frac{1}{2}\}$ is real.*

Proof. Let $t_0 \in \Omega$ satisfy $Z(t_0) = 0$. Theorem 3 gives $Y(\Theta(t_0))\sqrt{\Theta'(t_0)} = 0$. Lemma 4 gives $\sqrt{\Theta'(t_0)} \neq 0$, therefore $Y(\Theta(t_0)) = 0$. Corollary 2 gives $\Theta(t_0) \in \mathbb{R}$. Lemma 5, therefore, gives $t_0 \in \mathbb{R}$. $\square$

8 The Zero Locus of ζ

Corollary 3 (Riemann hypothesis)**.** *Every non-trivial zero ρ of the Riemann zeta function satisfies $\operatorname{Re} \rho = \frac{1}{2}$.*

Proof. Non-trivial zeros lie in $\{0 < \operatorname{Re} s < 1\}$, which under $s = \frac{1}{2} + it$ corresponds to Ω. For $\rho = \frac{1}{2} + it_0$ in Ω, $e^{-i\theta(t_0)}$ is non-vanishing and $\zeta(\frac{1}{2} + it_0) = e^{-i\theta(t_0)} Z(t_0)$, therefore $\zeta(\rho) = 0 \iff Z(t_0) = 0$. Theorem 4 gives $t_0 \in \mathbb{R}$, therefore $\operatorname{Re} \rho = \frac{1}{2}$. $\square$

A Answers to Frequently Anticipated Questions

This appendix collects short, colloquial explanations of terminology and conventions used in the main text.

A.1 Why "inverse phase pullback" and not "phase pushforward"?

The reason is semantic, not technical. The intuition is geometric, in the plain English sense of the words.

Pullback. Picture yourself standing at a point with stuff scattered across a distant landscape. You throw out strings, hook onto the stuff that is out there, and reel it back in toward where you are standing — like hauling in a kite. The object ends up concentrated at your location, retrieved from the space it was spread over. That retrieving motion is a pullback.

Pushforward. Now picture yourself sitting with a kite wound up at your feet and you want it out in the sky: you let out the line and send it forth, expanding it into the space it will occupy. That outgoing, spreading motion is a pushforward.

Apply that picture here. Z is being sent outward: it is spread across the t-line, expanded into that space. To build Y on the u-line, one sits at a point u, reaches back through the phase map via Θ^{-1}, grabs the value $Z(\Theta^{-1}(u))$, and reels it in. That retrieving is exactly why the operation is called a pullback: Z has been sent out, and one has to pull it back to gather it at u. Since the reeling is done along Θ^{-1}, the qualifier is "inverse phase." The adjective "unitary" records that the half-Jacobian $1/\sqrt{\Theta'(\Theta^{-1}(u))}$ is included so that L^2-mass is preserved under the operation.

Contravariance and covariance. The two directions carry standard names in differential geometry: a pullback is *contravariant* and a pushforward is *covariant*, referring to how each transforms under composition of maps. Objects that live as functions or scalar fields — things one evaluates at a point — transform contravariantly: they pull back, against the direction of the map. Objects that live as tangent vectors, velocities, or measures — things that carry a notion of flow or mass — transform covariantly: they push forward, with the direction of the map. The split is the same one that distinguishes covectors from vectors, or differential forms from densities: it records which kind of geometric object is being transported and which way the arrow of transport naturally runs. Under this classification, Z and Y are scalar-like, half-density data, and the natural transport between them is contravariant — a pullback.

A remark on the dual naming. Technically one could reverse the point of view and call Z itself the *phase pushforward* of the stationary process Y: starting from Y on the u-line, one lets out the line along Θ and expands Y outward into t-space, obtaining Z. Equivalently, the stationary process Y could be defined as the pushforward of Z along the phase — sending Z forth into u-coordinates. This naming is formally consistent, but it reads against the grain of what is being constructed. Stationarity is a property of something gathered and settled; producing it by *pushing* an object outward into a new space is the wrong English motion. One naturally assembles a stationary object by *pulling in* values from the non-stationary source. "Unitary inverse phase pullback" is therefore the name used throughout: it matches the geometric direction of the construction and the linguistic image of stationarity as something retrieved, not something broadcast.

Ditty.

Functions pull back, measures push forth.
Z lives in t, so u must go north:
compose with Θ^{-1}, put half-Jacobian in tow —
that's the unitary inverse phase pullback, row by row.

A.2 Why is the spectral measure non-negative?

The autocorrelation of any real, locally square-integrable function is a positive-definite tempered distribution: the Gram matrix of finitely many translates of Y restricted to a common window has non-negative eigenvalues, a one-line consequence of Cauchy–Schwarz. Bochner–Schwartz identifies the Fourier transform of a positive-definite tempered distribution with a non-negative tempered measure, and Wiener–Khinchin identifies that Fourier transform with $S = |\widehat{Y}|^2$. Non-negativity of the spectral measure is the Fourier-side restatement of this universal property of autocorrelations; the support condition $\operatorname{supp} S \subseteq [-1, 1]$ is delivered by Theorem 1 and combines with non-negativity to invoke Akhiezer.

A.3 What is being autocorrelated?

The autocorrelation in Lemma 9 is that of Y itself, the unitary inverse phase pullback of Z. In u-coordinates,

$$C(\eta) \;=\; \lim_{U\to\infty} \frac{1}{2U} \int_{-U}^{U} Y(u)\, Y(u+\eta)\, du,$$

a positive-definite tempered distribution. Because the map $Z \mapsto Y$ is an L^2-isometry under $u = \Theta(t)$, this is equivalent to correlating Z against itself with respect to the u-shift η, which is a nonlinear (roughly logarithmic) shift in the original t-coordinate.

A.4 What does the picture look like?

Three objects merit mental plots.

The function $Y(u)$. An envelope-modulated real oscillation of zero mean. The envelope decays like $1/\sqrt{\log t}$ expressed in u-coordinates. The zero crossings are the images under Θ of the nontrivial Riemann zeros; the phase variable straightens out their spacing.

The autocorrelation $C(\eta)$. A sharp peak at $\eta = 0$, symmetric in η because Y is real, with oscillatory decay whose frequency content sits inside $[-1, 1]$.

The spectral measure S. A non-negative tempered measure strictly supported in $[-1, 1]$, symmetric about $\xi = 0$ after the even extension. Its shape inside $[-1, 1]$ encodes the distribution of zero spacings of Z in u-coordinates.

A.5 Locally L^2 and the growing variance

Y is locally L^2, on compact subsets $[-U, U]$ of arbitrarily large length U. The band-limit established in Theorem 1 is a statement about the Fourier support of Y as a tempered distribution on $\mathbb{R}$. All uses of Paley–Wiener, Bochner–Schwartz, and Akhiezer in the main text are read in the tempered-distributional form, the framework for band-limited objects that are not globally L^2.

The growing variance manifests in three compatible ways.

Envelope in u*-coordinates.* $Y(u)$ has an amplitude envelope that grows slowly as $|u| \to \infty$; the local mean-square $\bar{P}(U) := \frac{1}{2U} \int_{-U}^{U} Y(u)^2 \, du$ grows without bound as $U \to \infty$.

Autocorrelation as a tempered distribution. $C(\eta)$ is the windowed-limit positive-definite distribution of §A.3. Bochner–Schwartz identifies its Fourier transform with the non-negative tempered measure S, and Theorem 1 localizes S to $[-1, 1]$.

Spectral measure as a tempered measure on $[-1, 1]$. S is a non-negative tempered measure on $\mathbb{R}$, supported in $[-1, 1]$, of infinite total mass.

Representation at finite window. Pointwise values of the spectral density at interior $\xi_0 \in (-1, 1)$ are approached through windowed periodograms

$$P_U(\xi_0) \;:=\; \frac{1}{2U} \left| \int_{-U}^{U} Y(u)\, e^{-i\xi_0 u} \, du \right|^2 .$$

Any figure produced at finite U is a *representation* at window scale U.

A.6 Scale of the spectral representation at finite window

Fix $\xi \in (-1, 1)$ and let $P_U(\xi)$ be the windowed periodogram. Two features of the plot change with U: the overall vertical scale, and the local fluctuation amplitude around that scale. Both are controlled by the local mean power

$$\bar{P}(U) \;:=\; \frac{1}{2U} \int_{-U}^{U} Y(u)^2 \, du \;=\; \frac{1}{2U} \int_{\Theta^{-1}(-U)}^{\Theta^{-1}(U)} Z(t)^2 \, dt,$$

the second equality by the L^2-isometry of the unitary inverse phase pullback. By Parseval in the tempered sense,

$$\int_{-\infty}^{\infty} P_U(\xi) \, d\xi \;=\; 2\pi \, \bar{P}(U).$$

Using $\Theta(t) \sim \frac{t}{2}\log(t/(2\pi e)) + ct$ and the Hardy–Littlewood moment $\int_0^T Z(t)^2\,dt \sim T\log T$ ([8], §7.3),

$$\bar{P}(U) \;\sim\; \log\bigl(\Theta^{-1}(U)\bigr) \;\sim\; \log U + \log\log U + O(1) \qquad (U \to \infty).$$

$P_U(\xi)$ fluctuates around $\bar{P}(U)$ on a scale comparable to $\bar{P}(U)$ itself, with wider fluctuations near the edges $|\xi| \to 1^-$.

A.7 The normalized spectral measure

At every finite U the periodogram P_U has finite total mass $2\pi\bar{P}(U)$ by Parseval. Dividing by this total mass produces the normalized spectral probability measure at scale U,

$$\widetilde{P}_U(\xi) \;:=\; \frac{P_U(\xi)}{2\pi\,\bar{P}(U)}, \qquad \int_{-\infty}^{\infty} \widetilde{P}_U(\xi)\,d\xi = 1,$$

a probability density on $\mathbb{R}$ at every U. By Theorem 1 the probability mass of $\widetilde{P}_U$ outside $[-1,1]$ vanishes as $U \to \infty$, so the family is tight on $[-1,1]$. By Helly–Prokhorov, $\widetilde{P}_U$ converges weakly as $U \to \infty$ to a probability measure ν_∞ on $[-1,1]$:

$$\int f\,d\widetilde{P}_U \;\longrightarrow\; \int f\,d\nu_\infty \qquad \text{for every } f \in C_b(\mathbb{R}).$$

ν_∞ is the normalized spectral density of Y: a probability measure of total mass 1 supported in $[-1,1]$.

S and ν_∞ are distinct, both well-defined:

- S: non-negative tempered spectral measure on $[-1,1]$, infinite total mass, delivered by Theorem 1 and Bochner–Schwartz ([7], Theorem 7.7).
- ν_∞: probability measure on $[-1,1]$, the weak-$*$ limit of the normalized windowed periodograms $\widetilde{P}_U$.

A.8 Cramér representation of the sample path

As a generalized stationary process on $\mathbb{R}_u$, Y admits a Cramér spectral representation of its sample path,

$$Y(u) \;=\; \int_{-1}^{1} e^{i\xi u}\,d\Phi(\xi),$$

where $d\Phi$ is a complex-valued orthogonal random measure on $[-1,1]$ with covariance structure

$$\mathbb{E}\bigl[d\Phi(\xi)\,\overline{d\Phi(\xi')}\bigr] \;=\; \delta(\xi-\xi')\,dS(\xi).$$

The support condition $\operatorname{supp} S \subseteq [-1,1]$ of Theorem 1 localizes $d\Phi$ to the interval $[-1,1]$. Reality of Y on $\mathbb{R}$ is equivalent to the Hermitian symmetry $d\Phi(-\xi) = \overline{d\Phi(\xi)}$, under which S is an even measure on $[-1,1]$. The integral is a stochastic integral in the orthogonal-increments sense; test functions f are paired with $d\Phi$ under the $L^2(S)$ norm identity

$$\mathbb{E}\Bigl|\int f(\xi)\,d\Phi(\xi)\Bigr|^2 \;=\; \int |f(\xi)|^2\,dS(\xi),$$

which requires $f \in L^2(S)$ and permits S to have infinite total mass. The Cramér representation exists at the level of S: it is the reconstruction theorem for the process. ν_∞ is the scale-invariant spectral shape of the same process.

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